

Abdul Wahid

Lab Report

Threat Intelligence Sharing – AlienVault OTX & IOC Bucket

Lab Overview

This lab demonstrates how to perform threat intelligence sharing using two platforms: AlienVault OTX (Open Threat Exchange) and IOC Bucket. The tasks include searching for malware file hash indicators, creating a pulse on AlienVault OTX, and uploading an IOC to IOC Bucket for community sharing.

Note: AlienVault OTX was used in place of ThreatConnect as specified in the lab manual, as it provides equivalent threat intelligence sharing capabilities with a free community account.

Task 1 – Threat Intelligence Sharing with AlienVault OTX

Step 1: AlienVault OTX Dashboard

Navigated to AlienVault OTX (otx.alienvault.com) and logged in. The dashboard displays a visualization of malware clusters, subscribed pulses from the community, and latest threat intelligence blogs from LevelBlue Labs.

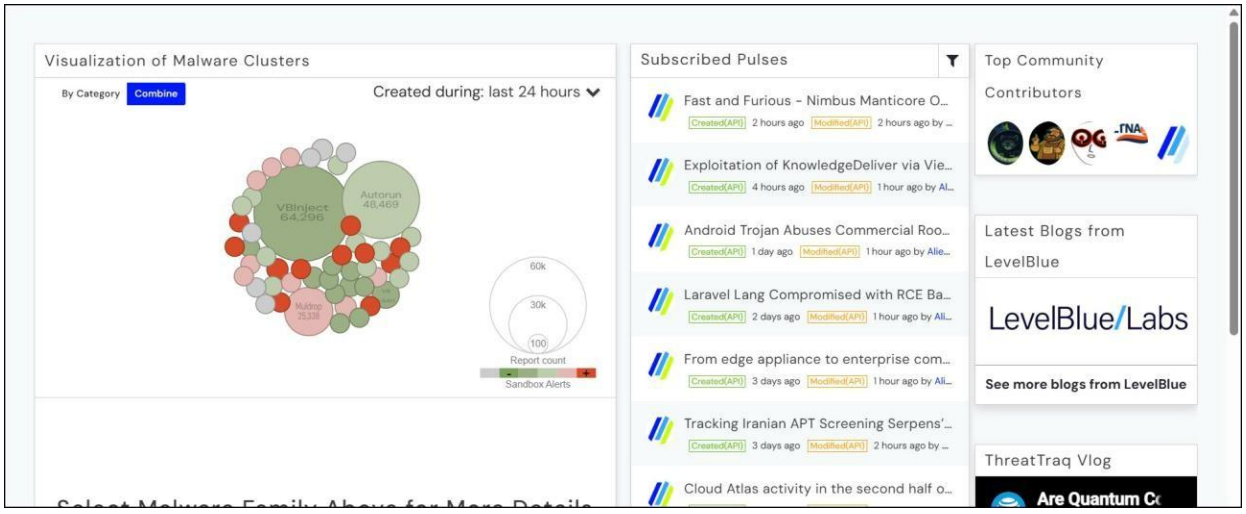


Figure 1.1: AlienVault OTX Dashboard – Malware Cluster Visualization and Subscribed Pulses

Step 2: Searching MD5 File Hash Indicator

Searched for the malware file hash indicator (MD5: f68872773a88652541bd8d6ca9bda058) on AlienVault OTX. The analysis overview shows the hash is referenced in 50 OTX User-Created Pulses with 13 related tags including Malware, YARA, Worm, MD5, SHA1, and SHA256.

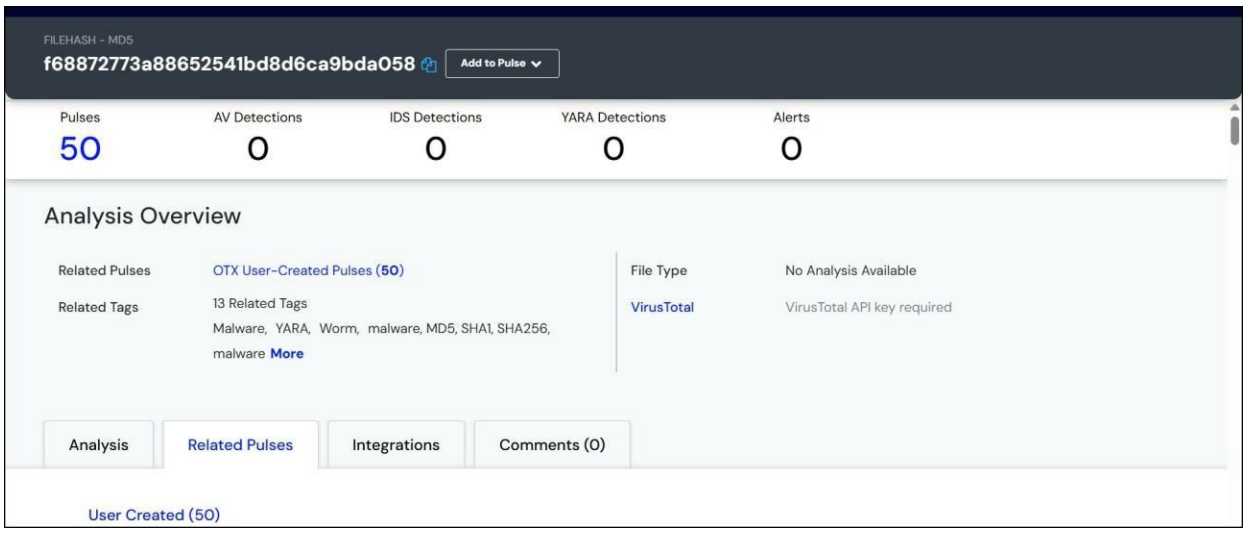


Figure 1.2: MD5 Hash Indicator Analysis – 50 Related Pulses Found

Step 3: Creating a New Pulse – Empty Form

Navigated to the "Create New Pulse" page on AlienVault OTX. The platform provides an extraction tool to automatically pull Indicators of Compromise (IOCs) from various sources including websites, PDF reports, emails, PCAP files, STIX, OpenIOC, CSV, or text files.

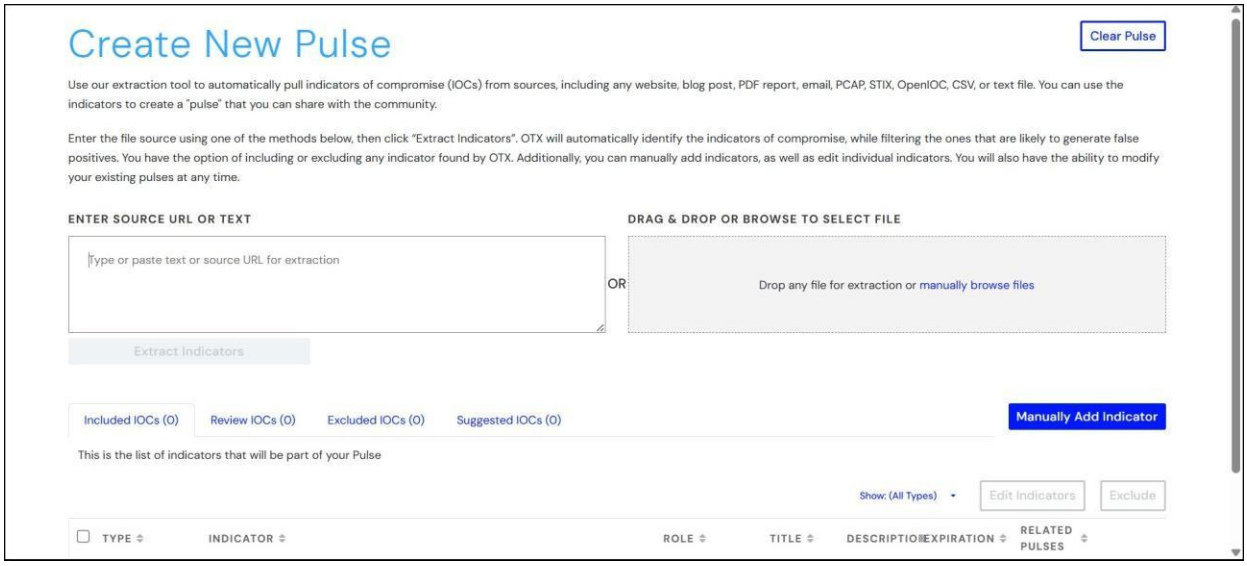


Figure 1.3: Create New Pulse – Empty Extraction Form

Step 4: Entering File Hash Indicators for Extraction

Entered the malware file hash indicators into the source text field for IOC extraction. The following hashes were entered:

- MD5: f68872773a88652541bd8d6ca9bda058
- SHA1: c8ac1b977d771c89ed7152f0daedb6a3e8b69b24
- SHA256: 9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7

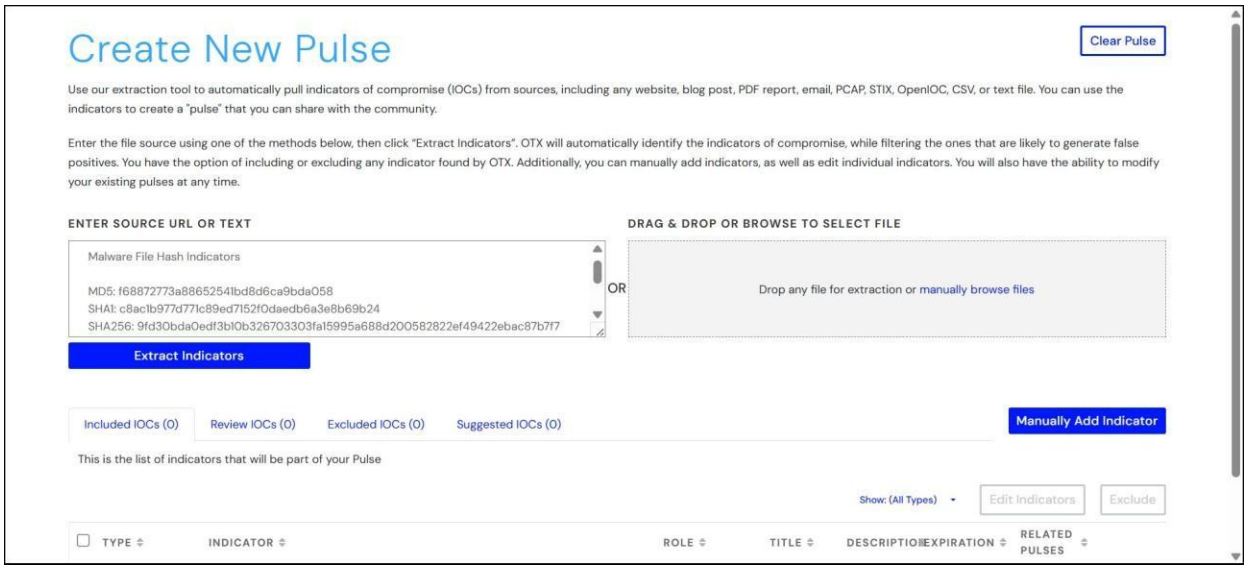


Figure 1.4: Pulse Creation – Entering MD5, SHA1, and SHA256 Hash Values

Step 5: Extracting IOC Indicators

Clicked the "Extract Indicators" button. OTX processed the input and automatically identified the indicators of compromise while filtering those likely to generate false positives.

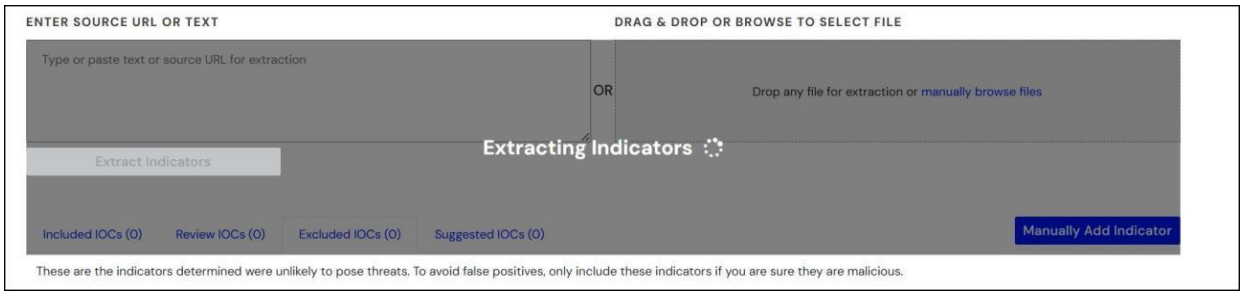


Figure 1.5: IOC Extraction in Progress

Step 6: Extracted IOCs – Included Indicators (3)

After extraction, OTX successfully identified 3 indicators: FileHash-MD5, FileHash-SHA1, and FileHash-SHA256. All three were placed in the "Included IOCs" tab, ready to be added to the pulse.



Figure 1.6: Included IOCs – Three File Hash Indicators Extracted Successfully

Step 7: Filling in Pulse Metadata

Filled in the pulse details including title, description, and tags, then reviewed the three included indicators before publishing:

- Title: Malware File Hash Indicators
- Description: This pulse contains MD5, SHA1, and SHA256 malware file hash indicators collected for threat intelligence sharing and analysis.
- TLP: Green (publicly shareable)
- Tags: hash, malware, malware file, sha1, sha256

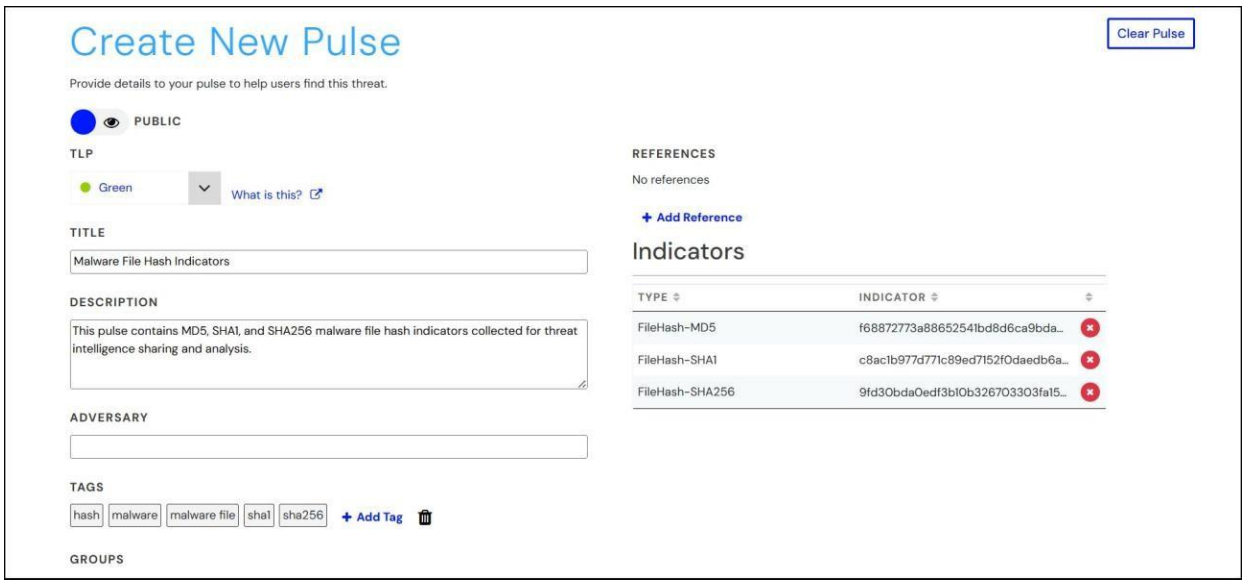


Figure 1.7: Pulse Metadata – Title, Description, Tags and Indicators

Step 8: Pulse Successfully Published

The pulse "Malware File Hash Indicators" was successfully created and published to the AlienVault OTX community. The pulse shows 3 Indicators of Compromise with one each of FileHash-MD5, FileHash-SHA1, and FileHash-SHA256, and 36 related pulses were automatically associated.

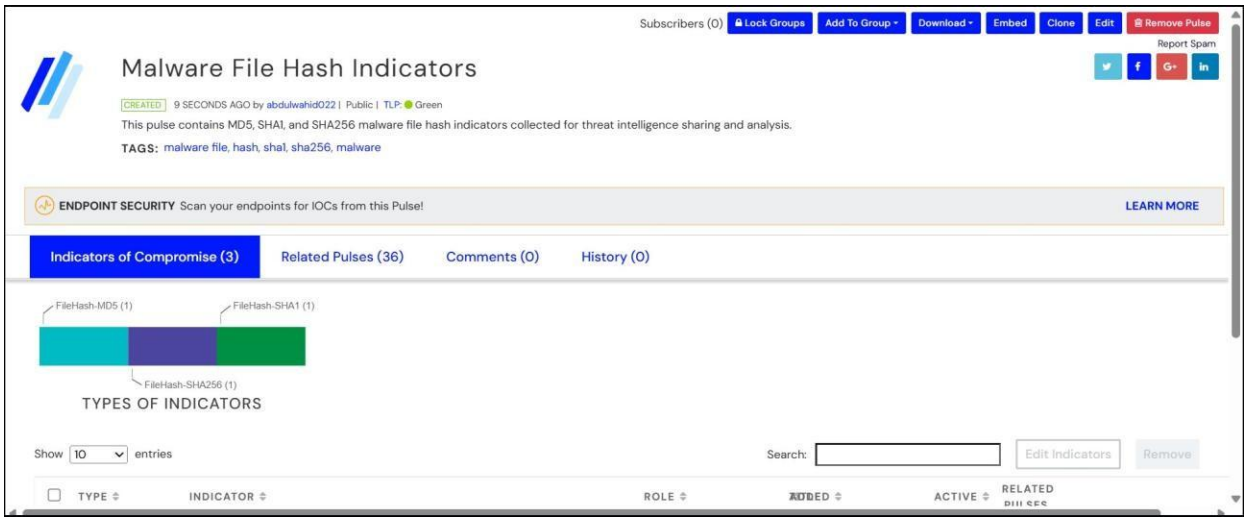


Figure 1.8: Published Pulse – Malware File Hash Indicators on AlienVault OTX

Task 2 – Threat Intelligence Sharing with IOC Bucket

Step 1: IOC Bucket Homepage

Navigated to https://www.iocbucket.com/ – a free community-driven platform dedicated to sharing quality threat intelligence. The platform currently indexes over 764 IOCs contributed by the security community.

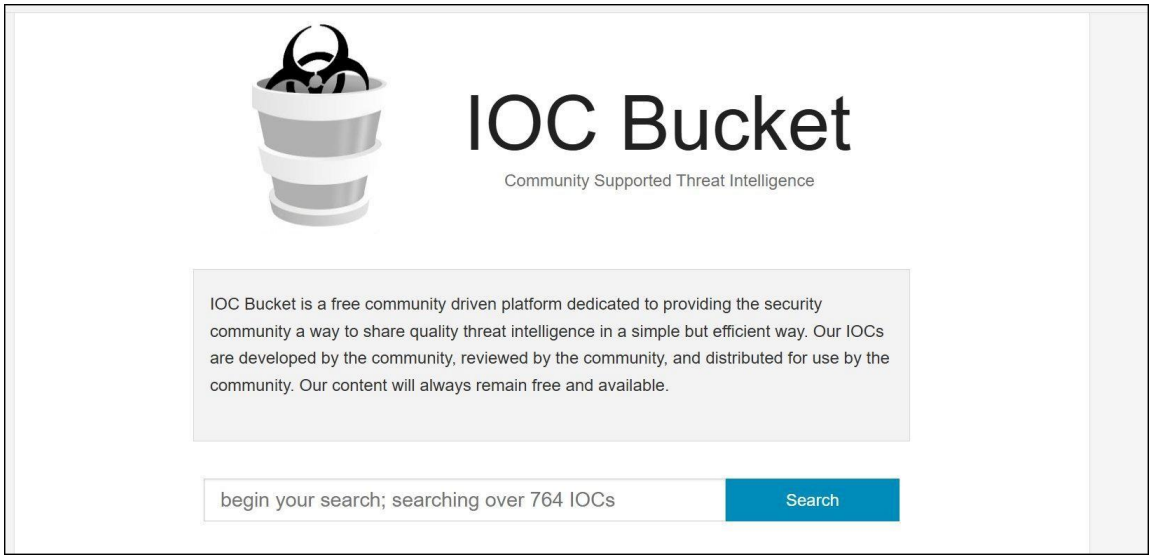


Figure 2.1: IOC Bucket Homepage – Community Supported Threat Intelligence Platform

Step 2: Navigating to the Upload Section

Clicked the "Upload" menu in the top navigation bar, which revealed two options: "Upload Single IOC" and "Upload OpenIOC in Bulk". Selected "Upload Single IOC" to upload the malware file hash indicator.

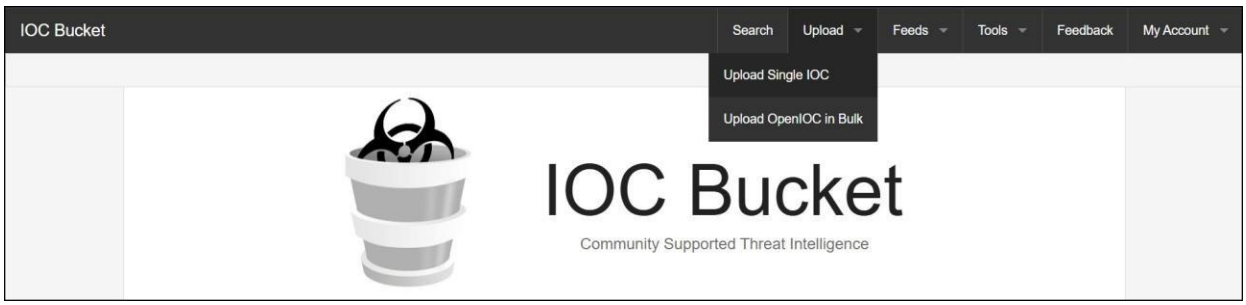


Figure 2.2: IOC Bucket Upload Menu – Single IOC and Bulk Upload Options

Step 3: Filling the Single Uploader Form

On the Single Uploader page, filled in all required details for the IOC submission:

- Author: Analyst Abdul Wahid
- Name: Malware File Hash IOC
- Short Description: Active malware file hash indicators detected in the system.
- Long Description: This IOC contains MD5, SHA1, and SHA256 malware file hash indicators for threat intelligence sharing and analysis.
- Tags: malware md5 sha1 sha256 filehash threat-intelligence

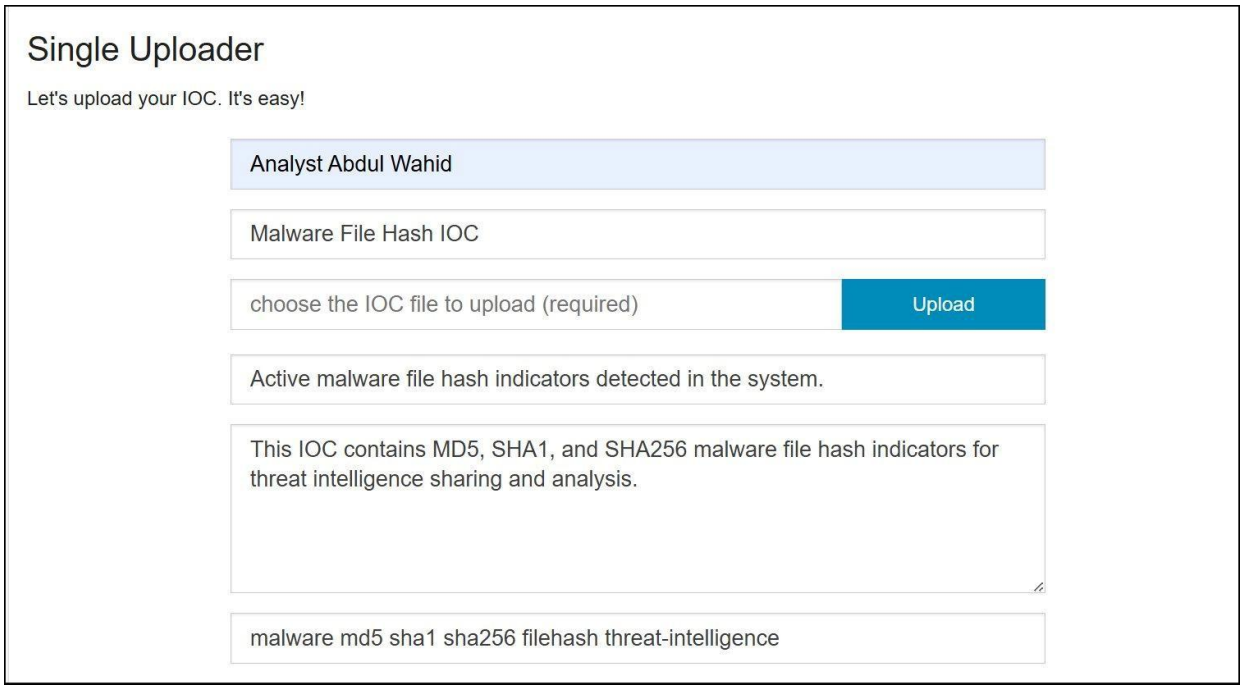


Figure 2.3: IOC Bucket Single Uploader Form – Details Filled In

Step 4: IOC Successfully Uploaded

Clicked the "Contribute!" button after filling in all details and selecting the IOC file. The platform displayed a success notification confirming the IOC was uploaded to the community feed.



Figure 2.4: Success Notification – "Your IOC was successfully uploaded."

Lab Analysis

This lab successfully demonstrated two approaches to threat intelligence sharing:

1. AlienVault OTX Pulse Creation: Using AlienVault OTX as an alternative to ThreatConnect, a new pulse titled "Malware File Hash Indicators" was created containing three IOCs (MD5, SHA1, SHA256 file hashes). The pulse was published publicly with TLP: Green classification, making it available to the entire OTX threat intelligence community.
2. IOC Bucket Submission: The malware file hash IOC was uploaded to IOC Bucket, a community-supported platform. The submission was confirmed with a success notification, contributing the indicator to the broader security community feed.

Both platforms demonstrate how threat analysts can share Indicators of Compromise with the security community, enabling proactive defense and collaborative threat detection.